

Project Title: Bluestock Mutual Fund Analysis & Dashboard Capstone

Name: VASALA SHIVANI

Date: July 4, 2026

Part 1: Daily Progress Summary

Day 1: Data Ingestion

- Successfully sourced the raw mutual fund datasets and imported them into the Python environment.
- Established the foundational data structure and verified file integrity.
- Performed an initial audit of data types and potential inconsistencies to map out the preprocessing steps.

Day 2: Data Cleaning

- Performed rigorous data cleaning, including the imputation of missing values and removal of duplicates.
- Standardized column naming conventions and reformatted date strings to ensure compatibility with SQL.
- Verified the cleaned dataset against original records to ensure no data loss occurred during transformation.

Day 3: Exploratory Data Analysis (EDA)

- Conducted statistical analysis on the cleaned data to understand distribution and performance outliers.
- Generated key visualizations, including time-series performance trends and volatility scatter plots.
- Derived actionable insights regarding correlation between fund categories and historical returns.

Day 4: Database Development

- Designed a normalized SQL database schema to effectively manage relationships between fund master and performance tables.
- Migrated the cleaned data into the structured relational database using Python-based connectors.
- Validated the schema by running internal integrity checks and testing basic primary-foreign key relationships.

Day 5: Dashboard Development

- Developed an interactive Power BI dashboard featuring drill-down capabilities for specific fund metrics.
- Integrated the SQL database as the backend data source, establishing live connection parameters.
- Implemented dynamic filters and slicers for date ranges and fund categories to improve user experience.

Day 6: Documentation & Optimization

- Drafted a comprehensive Data Dictionary documenting all features, units of measurement, and constraints.
- Refined Python scripts for database operations, adding error handling and logging for reproducibility.
- Reviewed code blocks to ensure compliance with PEP 8 standards and optimal execution speed.

Day 7: Final Packaging

- Structured the final project repository by organizing notebooks, scripts, and reports into a standardized directory.
- Finalized all documentation files, including README.md and requirements.txt, for project portability.
- Conducted a final quality check of all deliverables to ensure the submission meets all capstone requirements.